

Artificial Neural Networks

Feed- Forward and Back-Propagation Algorithm

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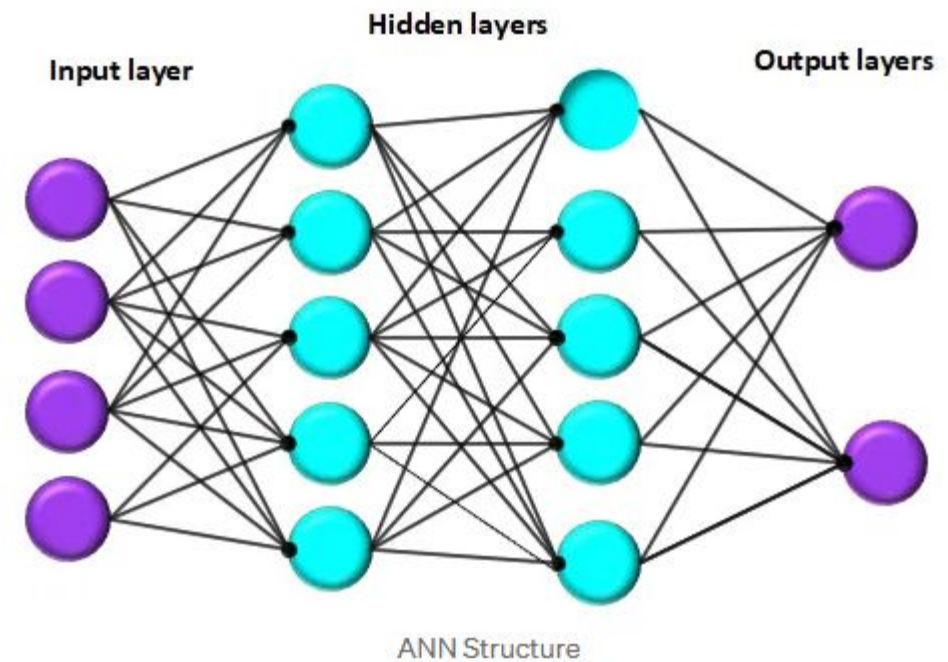
Artificial Neural Network (ANN)

- An **Artificial Neural Network (ANN)** is a computational model inspired by the **human brain**. It's designed to recognize patterns and relationships in data by mimicking how biological neurons signal and process information.

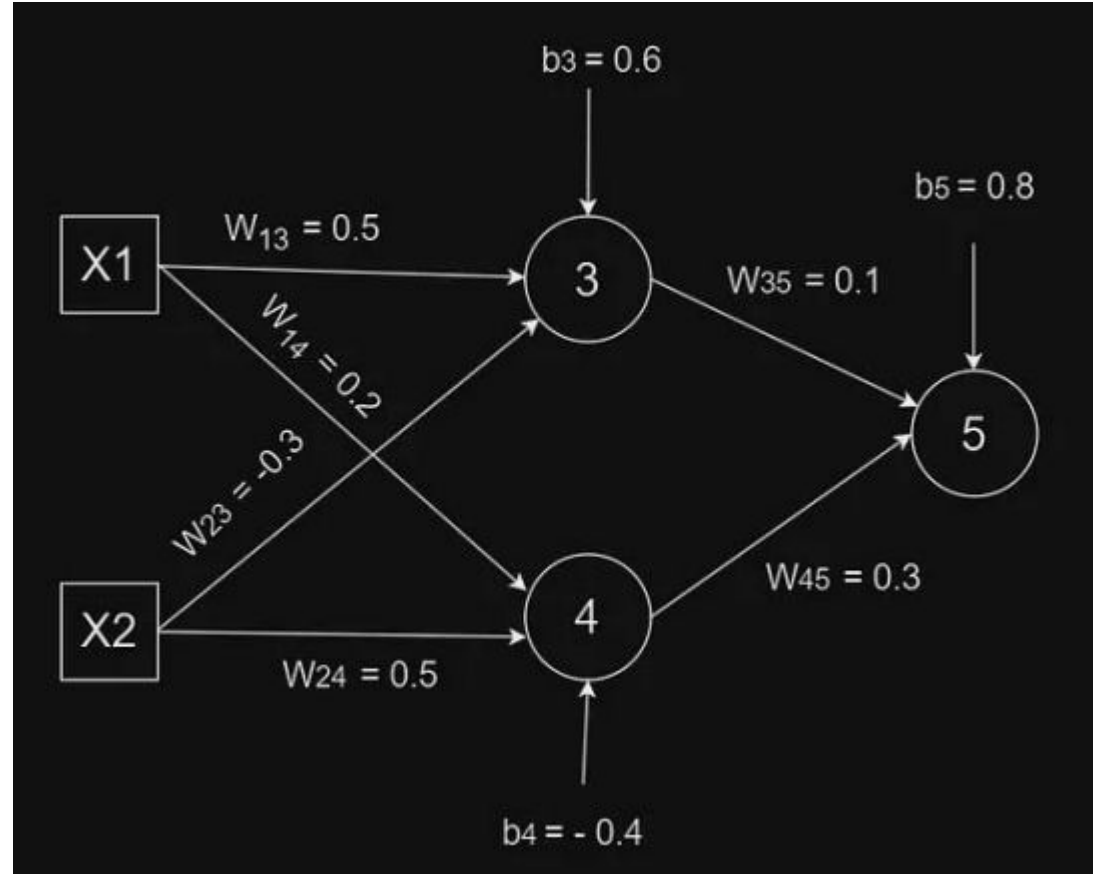
ANN Architecture Overview

An ANN consists of three main types of layers:

1. **Input Layer**: Receives the initial data (features).
2. **Hidden Layer(s)**: Processes data and extracts patterns (can be one or many layers).
3. **Output Layer**: Produces the final prediction (e.g., class label or value).
4. Each layer contains **neurons (nodes)** connected to neurons in the next layer via **weights**.



Network Architecture & Initial Values



Key Parameters:

- Learning rate (η) = 0.5
- Activation: Sigmoid
 $\sigma(x) = 1/(1+e^{-x})$
- Training tuples: (1,1,0) and (0,1,1)

Weight/Bias	Value
W_{13}	0.5
W_{14}	0.2
W_{23}	-0.3
W_{24}	0.5
W_{35}	0.1
W_{45}	0.3
b_3	0.6
b_4	-0.4
b_5	0.8

Initial Weights & Biases

Forward Pass — Iteration 1 | Input: ($X_1=1$, $X_2=1$, Target=0)

① INPUT LAYER

$$O_1 = I_1 = 1 \quad O_2 = I_2 = 1$$

② HIDDEN LAYER — Node 3

$$\begin{aligned} I_3 &= O_1 \times W_{13} + O_2 \times W_{23} + b_3 \\ &= 1 \times 0.5 + 1 \times (-0.3) + 0.6 \\ &= 0.5 - 0.3 + 0.6 = 0.8 \end{aligned}$$

$$\begin{aligned} O_3 &= \sigma(0.8) = 1/(1+e^{-0.8}) \\ O_3 &= 0.6900 \end{aligned}$$

② HIDDEN LAYER — Node 4

$$\begin{aligned} I_4 &= O_1 \times W_{14} + O_2 \times W_{24} + b_4 \\ \rightarrow &= 1 \times 0.2 + 1 \times 0.5 + (-0.4) \\ &= 0.2 + 0.5 - 0.4 = 0.3 \end{aligned}$$

$$\begin{aligned} O_4 &= \sigma(0.3) = 1/(1+e^{-0.3}) \\ O_4 &= 0.5744 \end{aligned}$$

③ OUTPUT LAYER — Node 5

$$\begin{aligned} I_5 &= O_3 \times W_{35} + O_4 \times W_{45} + b_5 \\ \rightarrow &= 0.69 \times 0.1 + 0.5744 \times 0.3 + 0.8 \\ &= 0.069 + 0.1723 + 0.8 \\ &= 1.0413 \end{aligned}$$

$$\begin{aligned} O_5 &= \sigma(1.0413) = 1/(1+e^{-1.0413}) \\ O_5 &= 0.7391 \end{aligned}$$

Predicted Output $O_5 = 0.7391$ | Target $\tau = 0$ | Error exists $\rightarrow$ Backpropagate!

Backpropagation — Error Calculation

Output Layer Formula: $Err_j = O_j (1 - O_j)(T_j - O_j)$

Hidden Layer Formula: $Err_j = O_j (1 - O_j) \times \sum Err_k \times w_{jk}$

ERROR AT OUTPUT NODE 5

$$\begin{aligned} Err_5 &= O_5 (1 - O_5)(T - O_5) \\ &= 0.7391 \times (1 - 0.7391) \times (0 - 0.7391) \\ &= 0.7391 \times 0.2609 \times (-0.7391) \\ &= -0.1425 \end{aligned}$$

ERROR AT HIDDEN NODE 4

$$\begin{aligned} Err_4 &= O_4(1-O_4)(Err_5 \times W_{45}) \\ &= 0.5744 \times (1-0.5744) \times (-0.1425 \times 0.3) \\ &= 0.5744 \times 0.4256 \times (-0.04275) \\ &= -0.010451 \end{aligned}$$

ERROR AT HIDDEN NODE 3

$$\begin{aligned} Err_3 &= O_3(1-O_3)(Err_5 \times W_{35}) \\ &= 0.69 \times (1-0.69) \times (-0.1425 \times 0.1) \\ &= 0.69 \times 0.31 \times (-0.01425) \\ &= -0.030481 \end{aligned}$$

Error Summary

$$\begin{aligned} Err_5 &= -0.1425 \\ Err_3 &= -0.030481 \\ Err_4 &= -0.010451 \end{aligned}$$

Weight & Bias Updates

Weight Update: $W_{ij}(\text{new}) = W_{ij}(\text{old}) + I \times O_i \times \text{Err}_j$

UPDATED WEIGHT VALUES

$W_{45}(\text{new})$	=	$0.3 + 0.5 \times 0.5744 \times (-0.1425)$	= 0.2591
$W_{35}(\text{new})$	=	$0.1 + 0.5 \times 0.69 \times (-0.1425)$	= 0.0508
$W_{24}(\text{new})$	=	$0.5 + 0.5 \times 1 \times (-0.010451)$	= 0.4948
$W_{14}(\text{new})$	=	$0.2 + 0.5 \times 1 \times (-0.010451)$	= 0.1948
$W_{23}(\text{new})$	=	$-0.3 + 0.5 \times 1 \times (-0.030481)$	= -0.3152
$W_{13}(\text{new})$	=	$0.5 + 0.5 \times 1 \times (-0.030481)$	= 0.4848

Bias Update: $b_j(\text{new}) = b_j(\text{old}) + I \times \text{Err}_j$

UPDATED BIAS VALUES

$$\begin{aligned} b_3 &= 0.6 + 0.5 \times (-0.030481) = 0.5848 \\ b_4 &= -0.4 + 0.5 \times (-0.010451) = -0.4052 \\ b_5 &= 0.8 + 0.5 \times (-0.1425) = 0.7288 \end{aligned}$$

Iteration 2 Forward Pass | Input: ($X_1=0$, $X_2=1$, Target=1)

Updated weights: $W_{13}=0.4848$ $W_{14}=0.1948$ $W_{23}=-0.3152$ $W_{24}=0.4948$ $W_{35}=0.0508$ $W_{45}=0.2591$ $b_3=0.5848$ $b_4=-0.4052$ $b_5=0.7288$

Hidden Node 3 ($X_1=0$, $X_2=1$)

$$\begin{aligned} I_3 &= 0 \times 0.4848 + 1 \times (-0.3152) + 0.5848 \\ &= 0 - 0.3152 + 0.5848 = 0.2696 \\ O_3 &= \sigma(0.2696) \approx 0.567 \end{aligned}$$

Hidden Node 4 ($X_1=0$, $X_2=1$)

$$\begin{aligned} I_4 &= 0 \times 0.1948 + 1 \times 0.4948 + (-0.4052) \\ &= 0 + 0.4948 - 0.4052 = 0.0896 \\ O_4 &= \sigma(0.0896) \approx 0.522 \end{aligned}$$

Output Node 5

$$\begin{aligned} I_5 &= O_3 \times W_{35} + O_4 \times W_{45} + b_5 = 0.567 \times 0.0508 + 0.522 \times 0.2591 + 0.7288 \\ &= 0.0288 + 0.1352 + 0.7288 = 0.8928 \\ O_5 &= \sigma(0.8928) \approx 0.710 \end{aligned}$$

Network is learning! Target=1 | O_5 improved toward target | Continue backpropagation with $\text{Err}_5 = O_5(1-O_5)(1-O_5)$

Iteration 2 Backpropagation — Error Calculation

$$\delta_3 = 0.5737(1 - 0.5737)(0.0598)(0.0509)$$

$$\delta_3 = 0.00075$$

$$\delta_4 = 0.5224(1 - 0.5224)(0.0598)(0.2591)$$

$$\delta_4 = 0.00386$$

- Target $t = 1$
- Output $O_5 = 0.710$

Formula:

$$\delta_5 = O_5(1 - O_5)(t - O_5)$$

Substitution:

$$\delta_5 = 0.710(1 - 0.710)(1 - 0.710)$$

$$= 0.710(0.290)(0.290)$$

$$= 0.710 \times 0.0841$$

$$\delta_5 \approx 0.0597$$

Iteration 2 Weight & Bias Updates

Hidden → Output

$$W_{35} = 0.0509 + 0.5(0.0598)(0.5737) = 0.0680$$

$$W_{45} = 0.2591 + 0.5(0.0598)(0.5224) = 0.2747$$

Input → Hidden

$$W_{13} = 0.4985 + 0.5(0.00075)(0) = 0.4985$$

$$W_{23} = -0.3015 + 0.5(0.00075) = -0.3011$$

$$W_{14} = 0.1948 + 0.5(0.00386)(0) = 0.1948$$

$$W_{24} = 0.4948 + 0.5(0.00386) = 0.4967$$

Weight/Bias	Value
W13	0.4985
W23	-0.3011
W14	0.1948
W24	0.4967
W35	0.0680
W45	0.2747
b3	0.5989
b4	-0.4033
b5	0.7587

Update Biases

$$b_5 = 0.7288 + 0.5(0.0598) = 0.7587$$

$$b_3 = 0.5985 + 0.5(0.00075) = 0.5989$$

$$b_4 = -0.4052 + 0.5(0.00386) = -0.4033$$

- Positive error means output is lower than target.
- Network will increase weights contributing positively to output.
- This helps push **O₅** closer to 1 in future iterations.

Core Comparison

Feature	Logistic Regression	ANN
Layers	Single layer	Multiple layers
Decision Boundary	Linear	Nonlinear
Training	Gradient descent	Backpropagation + gradient descent
Speed	Fast	Slower
Interpretability	High	Low
Data Requirement	Small-medium	Large
Complexity	Low	High

Choose Logistic Regression when:

- Problem is simple
- Data is linearly separable
- Interpretability matters
- Quick solution needed

Choose ANN when:

- Problem is complex
- Nonlinear relationships exist
- Large dataset available
- Higher predictive power is needed

Summary & Key Takeaways



Algorithm Steps

1. Initialize weights & biases
2. Feed-forward: compute outputs
3. Calculate error at output
4. Backpropagate error to hidden layers
5. Update all weights & biases
6. Repeat until convergence



Key Formulas

Output error: $O_j - T_j$
 $(1 - O_j)(T_j - O_j)$

Hidden error: $O_j(1 - O_j) \sum \text{Err}_k w_{kj}$

Weight update: $W_{ij} + \eta \times O_i \times \text{Err}_j$

Bias update: $b_j + \eta \times \text{Err}_j$

Sigmoid: $\sigma(x) = 1/(1 + e^{-x})$



Iteration 1 Results

Initial $O_5 = 0.7391$ (target = 0)

$\text{Err}_5 = -0.1425$

After update: $O_5 = 0.710$

New $\text{Err}_5 = 0.0597$

Network is converging ✓

Training continues iterating over all tuples until the global error is minimized (convergence)